NON - PARAMETRIC TESTS

Dr. Pramila Chaubey

Contents

- Introduction
- Assumptions of parametric and non-parametric tests
- Testing the assumption of normality
- Commonly used non-parametric tests
- Applying tests in SPSS
- Advantages of non-parametric tests
- Limitations
- Summary

Introduction

- Variable: A characteristic that is observed or manipulated.
  - **Dependent**
  - **Independent**

- Data: Measurements or observations of a variable
  1. **Nominal or Classificatory Scale:** Gender, ethnic background, eye colour, blood group
  2. **Ordinal or Ranking Scale:** School performance, social economic class
  3. **Interval Scale:** Celsius or Fahrenheit scale
  4. **Ratio Scale:** Kelvin scale, weight, pulse rate, respiratory rate

Introduction

- **Parameter**: is any numerical quantity that characterizes a given population or some aspect of it. Most common statistics parameters are mean, median, mode, standard deviation.

Assumptions

- The **general assumptions** of parametric tests are
  - The populations are **normally distributed** (follow normal distribution curve).
  - The selected population is **representative of general population**
  - The data is in **interval or ratio scale**

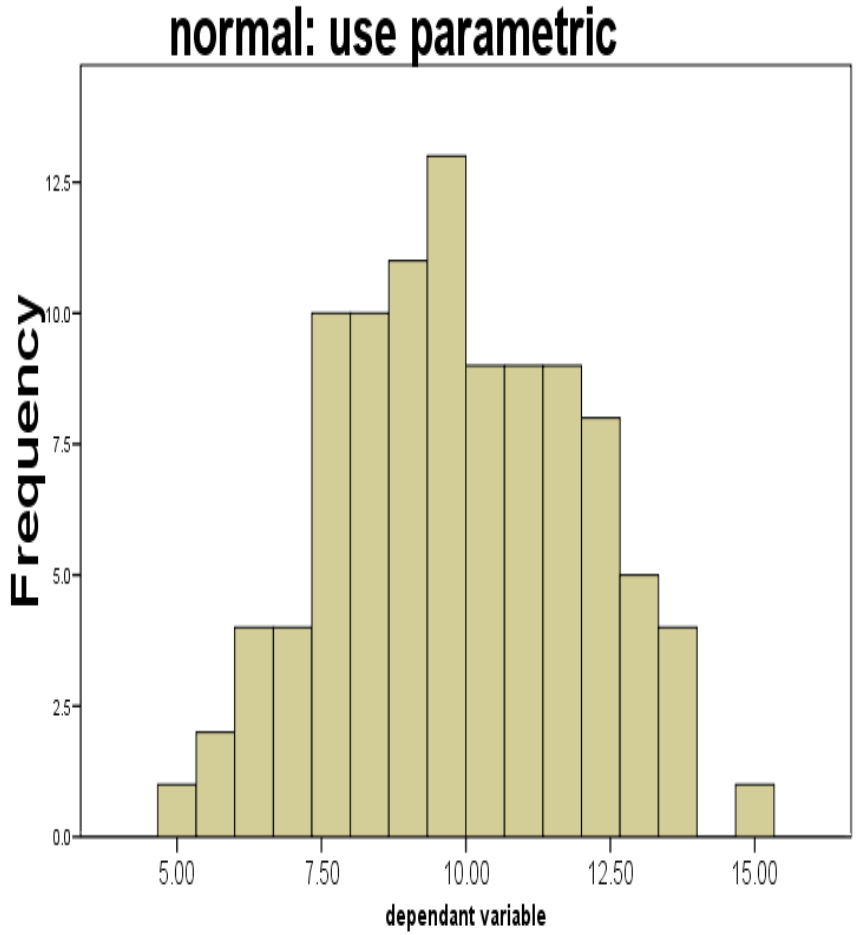
Assumptions

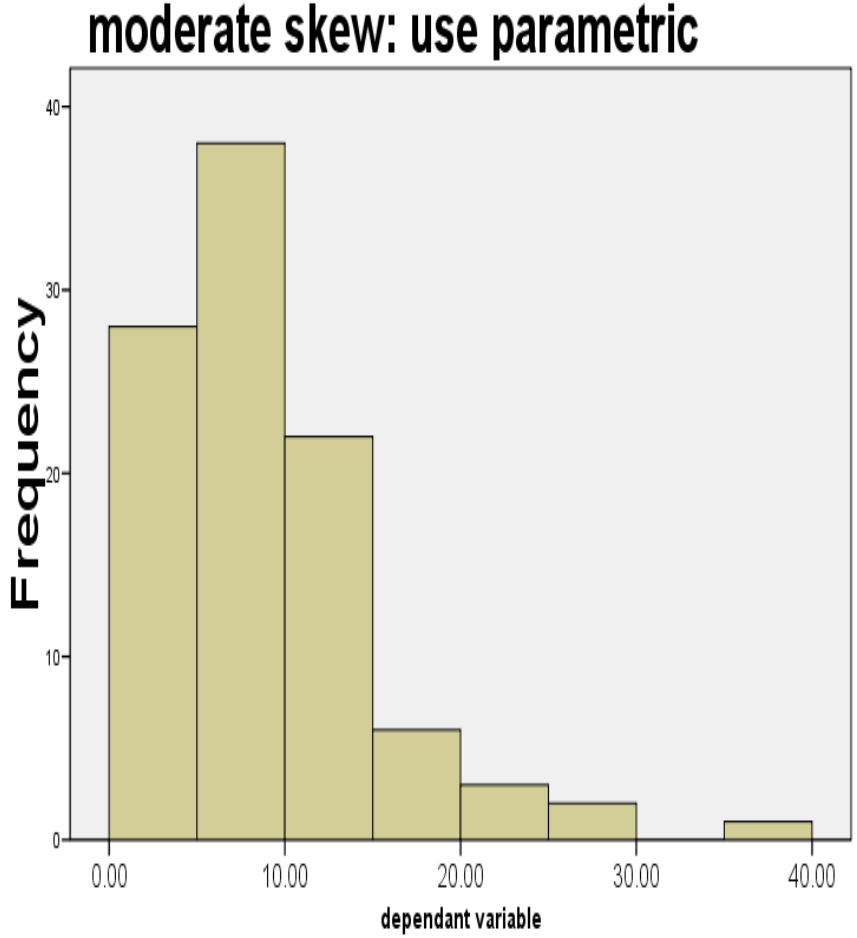
- Non-parametric tests can be applied when:
  - Data don't follow any specific distribution and no assumptions about the population are made
  - Data measured on any scale

Testing normality

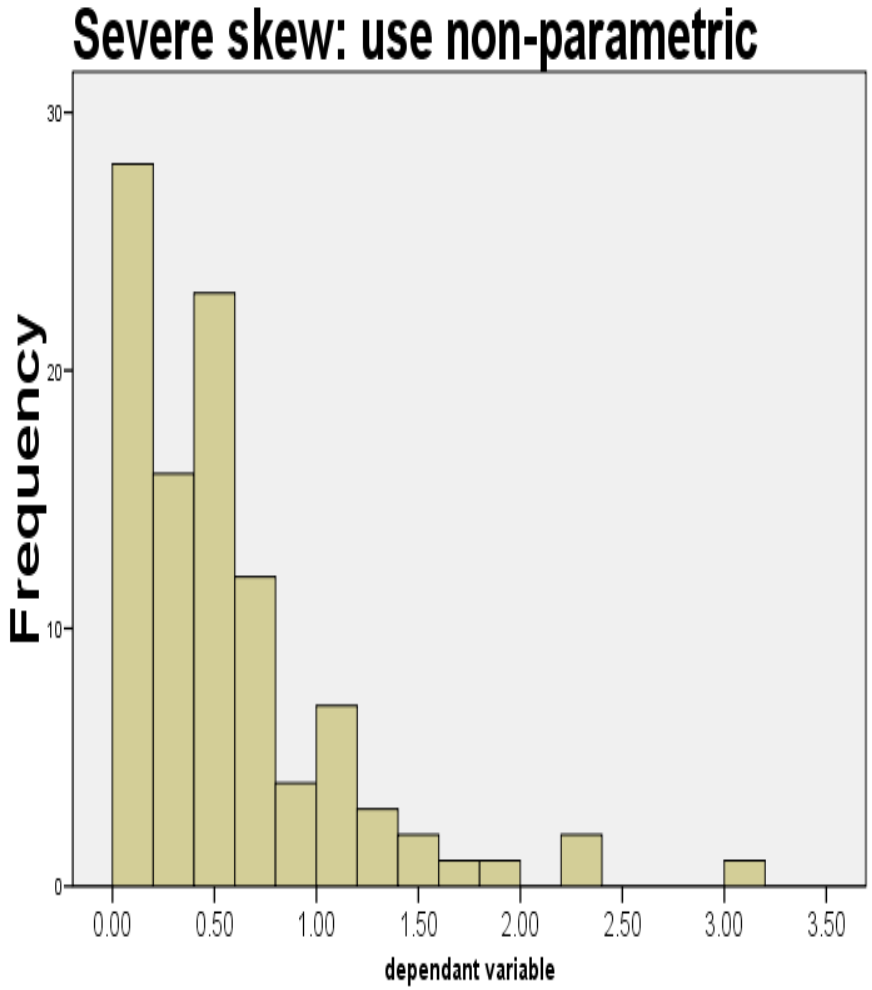
- Normality: This assumption is only broken if there are large and obvious departures from normality
- This can be checked by
  - ✓ Inspecting a histogram
  - ✓ Skewness and kurtosis **(** **K**urtosis describes the **pea**k of the curve **S**kewness describes the **S**ymmetry of the curve**.)**
  - ✓ Kolmogorov-Smirnov (K-S) test (sample size is ≥50)
  - ✓ Shapiro-Wilk test (if sample size is <50)
    (Sig. value >0.05 indicates normality of the distribution)

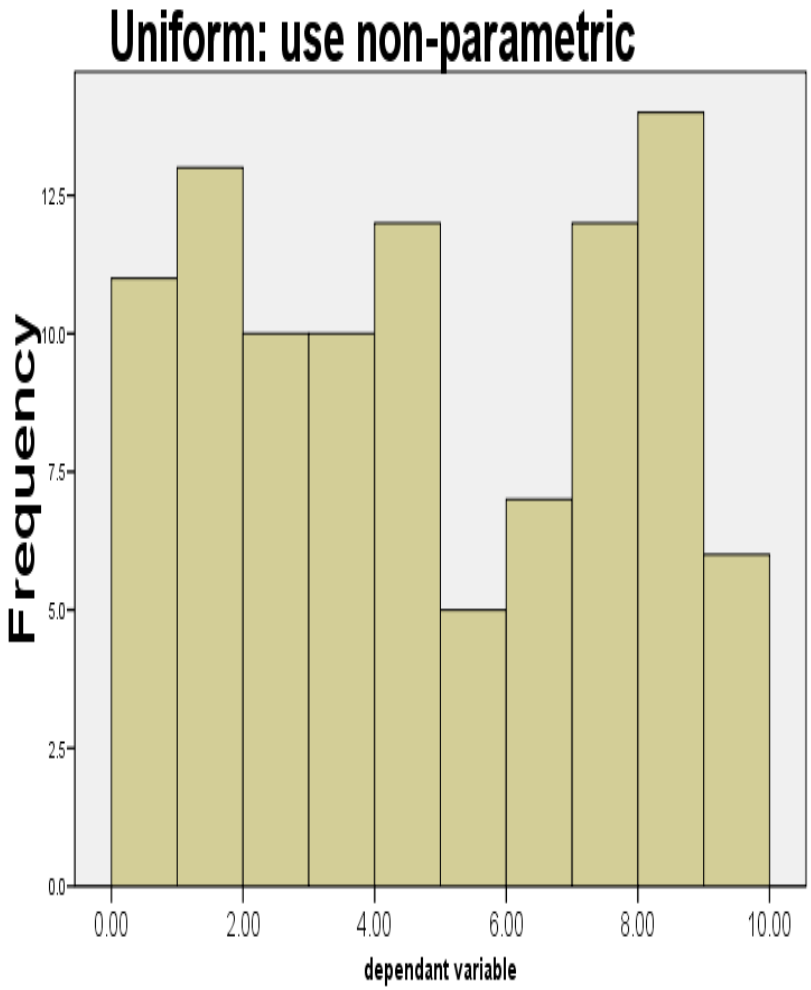
Testing normality



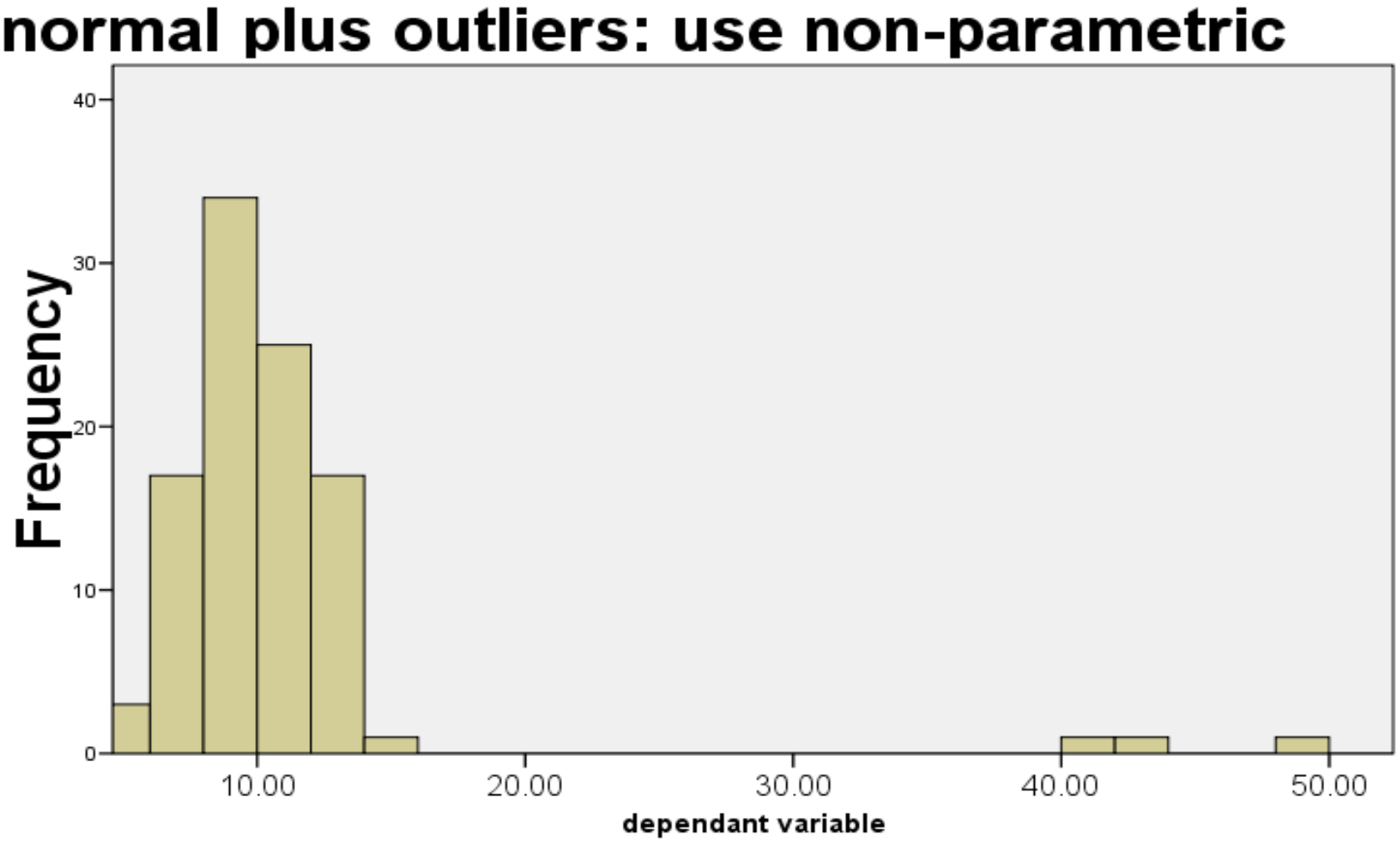


Testing normality





Testing normality



Commonly used tests

- Commonly used Non Parametric Tests are:
  - Chi Square test
  - McNemar test
  - The Sign Test
  - Wilcoxon Signed-Ranks Test
  - Mann–Whitney U or Wilcoxon rank sum test
  - The Kruskal Wallis or H test
  - Friedman ANOVA
  - The Spearman rank correlation test
  - Cochran's Q test

Chi Square test

- First used by Karl Pearson
- Simplest & most widely used non-parametric test in statistical work.
- Calculated using the formula-

$$\chi 2 = \sum \frac{(O - E)^2}{E}$$

O = observed frequencies

E = expected frequencies



Karl Pearson (1857–1936)

- Greater the discrepancy b/w observed & expected frequencies, greater shall be the value of $\chi 2$.
- Calculated value of $\chi 2$ is compared with table value of $\chi 2$ for given degrees of freedom.

Chi Square test

- Application of chi-square test:
  - Test of association (smoking & cancer, treatment & outcome of disease, vaccination & immunity)
  - Test of proportions (compare frequencies of diabetics & non-diabetics in groups weighing 40-50kg, 50-60kg, 60-70kg & >70kg.)
  - The chi-square for goodness of fit (determine if actual numbers are similar to the expected/theoretical numbers)

Chi Square test

- Attack rates among vaccinated & unvaccinated children against measles :

Group	Result		Total
	Attacked	Not-attacked	
Vaccinated (observed)	10	90	100
Unvaccinated (observed)	26	74	100
Total	36	164	200

- Prove protective value of vaccination by $\chi 2$ test at 5% level of significance

Chi Square test

Group	Result		Total
	Attacked	Not-attacked	
Vaccinated (Expected)	18	82	100
Unvaccinated (Expected)	18	82	100
Total	36	164	200

Chi Square test

- ✓ χ^2 value $= \sum (O-E)^2/E$
- ✓ $\frac{(10-18)^2}{18} + \frac{(90-82)^2}{82} + \frac{(26-18)^2}{18} + \frac{(74-82)^2}{82}$
- ✓ $\frac{64}{18} + \frac{64}{82} + \frac{64}{18} + \frac{64}{82}$
- ✓ $=8.67$
- ✓ calculated value (8.67) > 3.84 (expected value corresponding to P=0.05)
- ✓ **Null hypothesis is rejected. Vaccination is protective.**

Chi Square test

- **Yates' correction:** applies when we have two categories (one degree of freedom)
- Used when sample size is ≥ 40, and expected frequency of <5 in one cell
- Subtracting 0.5 from the difference between each observed value and its expected value in a 2 × 2 contingency table

- $$\chi^2 = \sum \frac{\lceil O - E - 0.5 \rceil^2}{E}$$

Fisher's Exact Test

- Used when the
  - Total number of cases is <20 or
  - The expected number of cases in any cell is ≤1 or
  - More than 25% of the cells have expected frequencies <5.



Ronald A. Fisher (1890–1962)

McNemar Test

- **McNemar Test:** used to compare before and after findings in the same individual or to compare findings in a matched analysis (for dichotomous variables)
- **Example**: comparing the attitudes of medical students toward confidence in statistics analysis before and after the intensive statistics course.



McNemar

Sign Test

- Used for paired data, can be ordinal or continuous
- Simple and easy to interpret
- Makes no assumptions about distribution of the data
- Not very powerful
- To evaluate H_0 we only need to know the signs of the differences
- If half the differences are positive and half are negative, then the median = 0 (H_0 is true).
- If the signs are more unbalanced, then that is evidence against H_0.

Sign Test

- Children in an orthodontia study were asked to rate how they felt about their teeth on a 5 point scale.
- Survey administered before and after treatment.

How do you feel about your teeth?

1. Wish I could change them
2. Don't like, but can put up with them
3. No particular feelings one way or the other
4. I am satisfied with them
5. Consider myself fortunate in this area

child	Rating before	Rating after
1	**1**	**5**
2	**1**	**4**
3	**3**	**1**
4	**2**	**3**
5	**4**	**4**
6	**1**	**4**
7	**3**	**5**
8	**1**	**5**
9	**1**	**4**
10	**4**	**4**
11	**1**	**1**
12	**1**	**4**
13	**1**	**4**
14	**2**	**4**
15	**1**	**4**
16	**2**	**5**
17	**1**	**4**
18	**1**	**5**
19	**4**	**4**
20	**3**	**5**

- Use the **sign test** to evaluate whether these data provide evidence that orthodontic treatment improves children's image of their teeth.

child	Rating before	Rating after	change
1	1	5	4
2	1	4	3
3	3	1	-2
4	2	3	1
5	4	4	0
6	1	4	3
7	3	5	2
8	1	5	4
9	1	4	3
10	4	4	0
11	1	1	0
12	1	4	3
13	1	4	3
14	2	4	2
15	1	4	3
16	2	5	3
17	1	4	3
18	1	5	4
19	4	4	0
20	3	5	2

- First, for each child, compute the difference between the two ratings

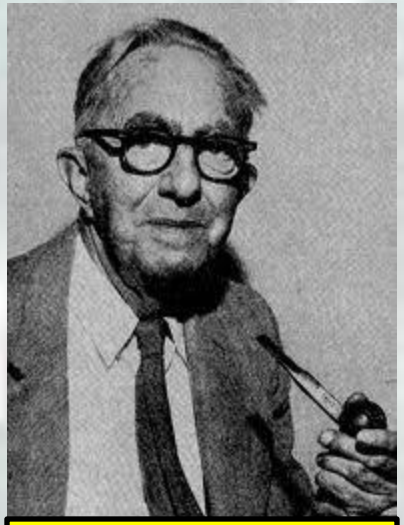
child	Rating before	Rating after	change	sign
1	1	5	4	+
2	1	4	3	+
3	3	1	-2	-
4	2	3	1	+
5	4	4	0	0
6	1	4	3	+
7	3	5	2	+
8	1	5	4	+
9	1	4	3	+
10	4	4	0	0
11	1	1	0	0
12	1	4	3	+
13	1	4	3	+
14	2	4	2	+
15	1	4	3	+
16	2	5	3	+
17	1	4	3	+
18	1	5	4	+
19	4	4	0	0
20	3	5	2	+

- The sign test looks at the signs of the differences
  - **15** children felt better about their teeth (+ difference in ratings)
  - **1** child felt worse (- diff.)
  - **4** children felt the same (difference = 0)
- If H_0 were true we'd expect an equal number of positive and negative differences.

  (P value from table 0.004)

Wilcoxon signed-rank test

- Nonparametric equivalent of the paired t-test.
- Similar to sign test, but take into consideration the magnitude of difference among the pairs of values. (Sign test only considers the direction of difference but not the magnitude of differences.)



WILCOXON

Wilcoxon signed-rank test

- The 14 difference scores in BP among hypertensive patients after giving drug A were:

  -20, -8, -14, -12, -26, +6, -18, -10, -12, -10, -8, +4, +2, -18

- The statistic T is found by calculating the sum of the positive ranks, and the sum of the negative ranks.
- The smaller of the two values is considered.

Wilcoxon signed-rank test

Score	Rank
+2	1
+4	2
+6	3
-8	4.5
-8	4.5
-10	6.5
-10	6.5
-12	8
-14	9
-16	10
-18	11.5
-18	11.5
-20	13
-26	14

Sum of positive ranks = 6

Sum of negative ranks = 99

T= 6

For N = 14, and $\alpha = .05$, the critical value of T = 21.
If T is equal to or less than T critical, then null hypothesis is rejected i.e., drug A decreases the BP among hypertensive patients.

Mann-Whitney U test

- Mann-Whitney U – similar to Wilcoxon signed-ranks test except that the samples are independent and not paired.
- Null hypothesis: the population means are the same for the two groups.
- Rank the combined data values for the two groups. Then find the average rank in each group.

Mann-Whitney U test

- Then the U value is calculated using formula
- $U = N1*N2 + \frac{Nx(Nx+1)}{2} - Rx$ (where Rx is larger rank total)
- To be statistically significant, obtained U has to be equal to or LESS than this critical value.

Mann-Whitney U test

- 10 dieters following Atkin's diet vs. 10 dieters following Jenny Craig diet

- Hypothetical RESULTS:
- Atkin's group loses an average of 34.5 lbs.

- J. Craig group loses an average of 18.5 lbs.

- Conclusion: Atkin's is better?

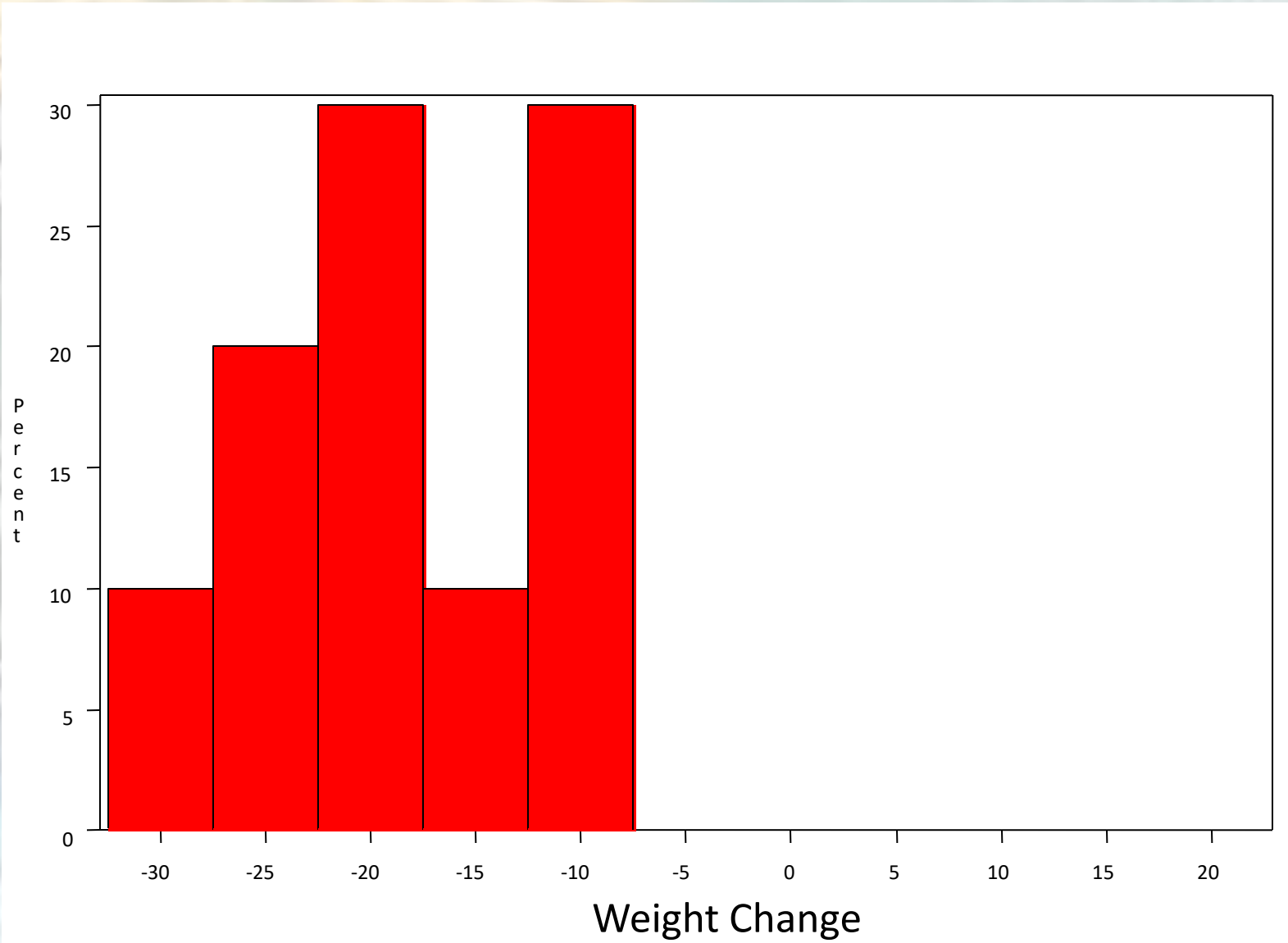
Mann-Whitney U test

- When individual data is seen
- Atkin's, change in weight (lbs):

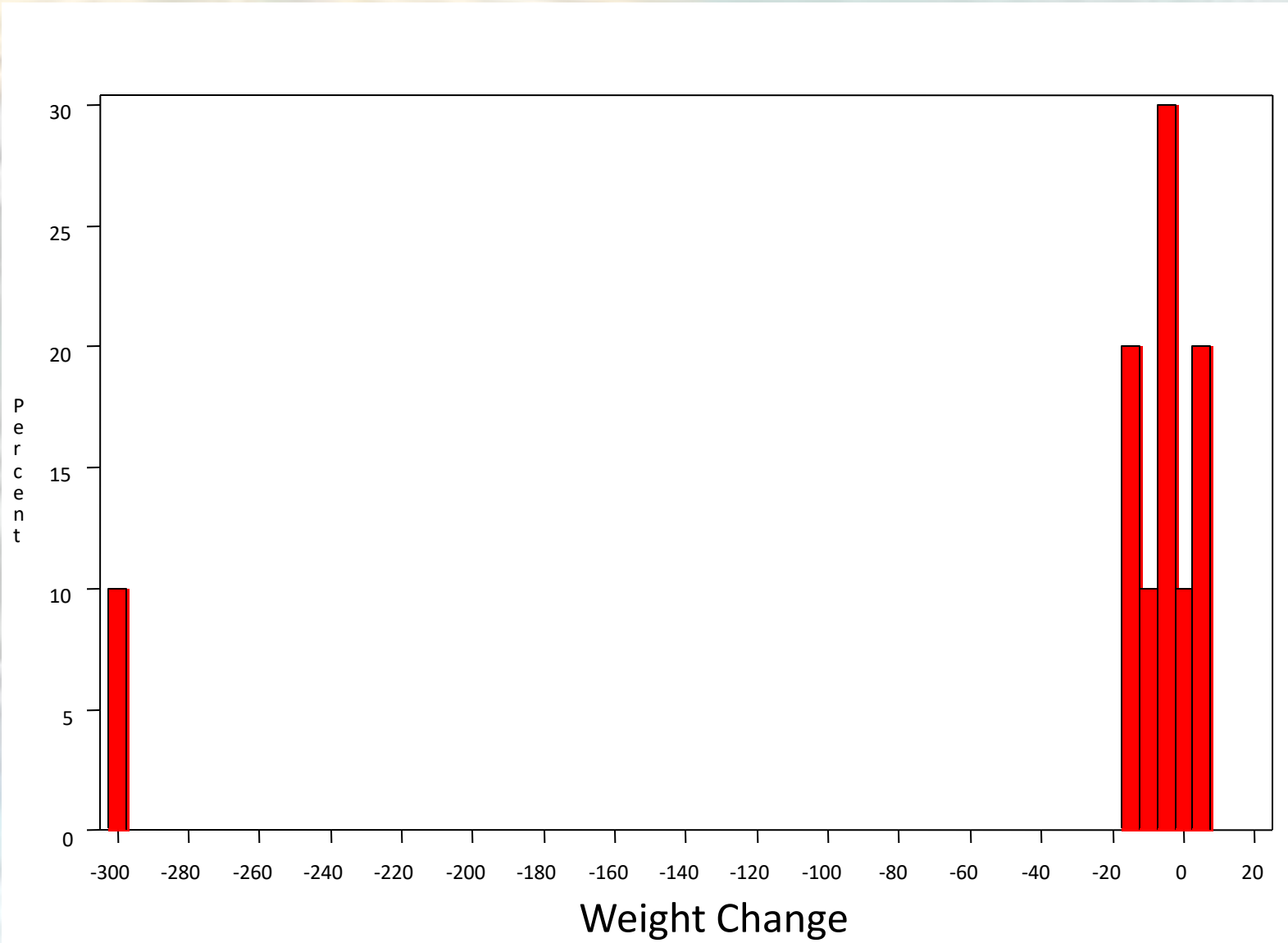
  +4, +3, 0, -3, -4, -5, -11, -14, -15, -300
- J. Craig, change in weight (lbs)

  -8, -10, -12, -16, -18, -20, -21, -24, -26, -30

Jenny Craig diet



Atkin's diet


30
25
20
15
10
5
0
Percent
-300
-280
-260
-240
-220
-200
-180
-160
-140
-120
-100
-80
-60
-40
-20
0
20
Weight Change

Mann-Whitney U test

- RANK the values, 1 being the least weight loss and 20 being the most weight loss.
- Atkin's
  - +4, +3, 0, -3, -4, -5, -11, -14, -15, -300
  - 1, 2, 3, 4, 5, 6, 9, 11, 12, 20
- J. Craig
  - -8, -10, -12, -16, -18, -20, -21, -24, -26, -30
  - 7, 8, 10, 13, 14, 15, 16, 17, 18, 19

Mann-Whitney U test

- Sum of Atkin's ranks:

  1+ 2 + 3 + 4 + 5 + 6 + 9 + 11+ 12 + 20=73

- Sum of Jenny Craig's ranks:

  7 + 8 +10+ 13+ 14+ 15+16+ 17+ 18+19=137

- Jenny Craig clearly ranked higher.
- Calculated U value (18) < table value (27), Null hypothesis is rejected.

Kruskal-Wallis One-way ANOVA

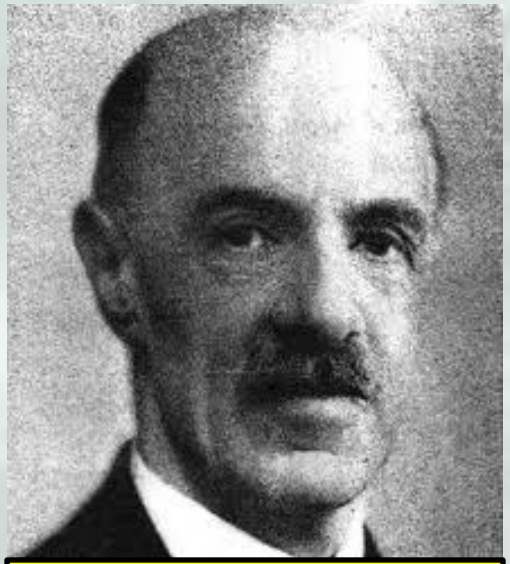
- It’s more powerful than Chi-square test.
- It is computed exactly like the Mann-Whitney test, except that there are more groups (>2 groups).
- Applied on independent samples with the same shape (but not necessarily normal).

Friedman ANOVA

- Friedman ANOVA: When either a matched-subjects or repeated-measure design is used and the hypothesis of a difference among three or more (k) treatments is to be tested, the Friedman ANOVA by ranks test can be used.

Spearman rank-order correlation

- Use to assess the relationship between two ordinal variables or two skewed continuous variables.
- Nonparametric equivalent of the Pearson correlation.
- It is a relative measure which varies from -1 (perfect negative relationship) to +1 (perfect positive relationship).



Charles Spearman
(1863–1945)

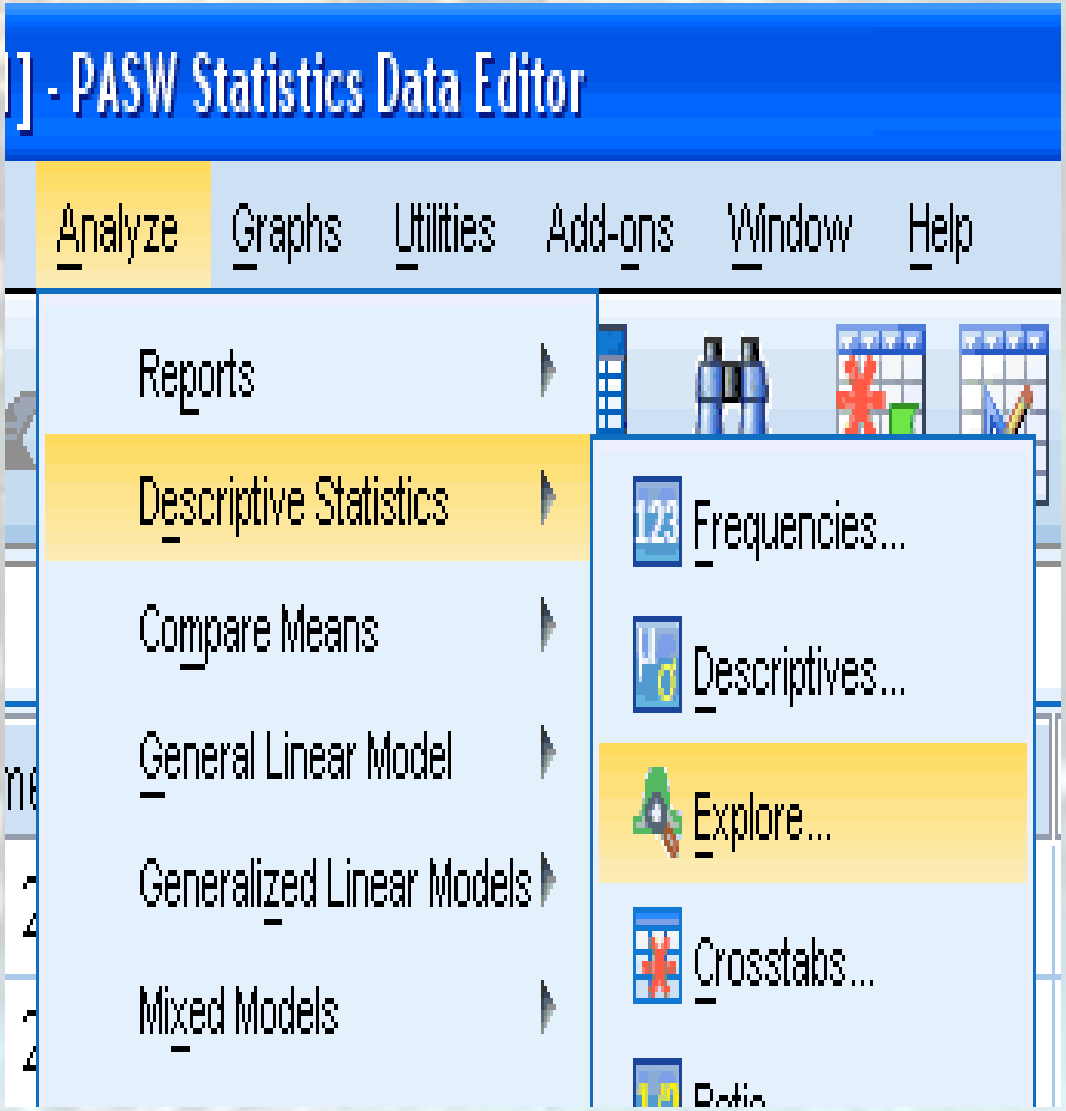
Cochran's Q test

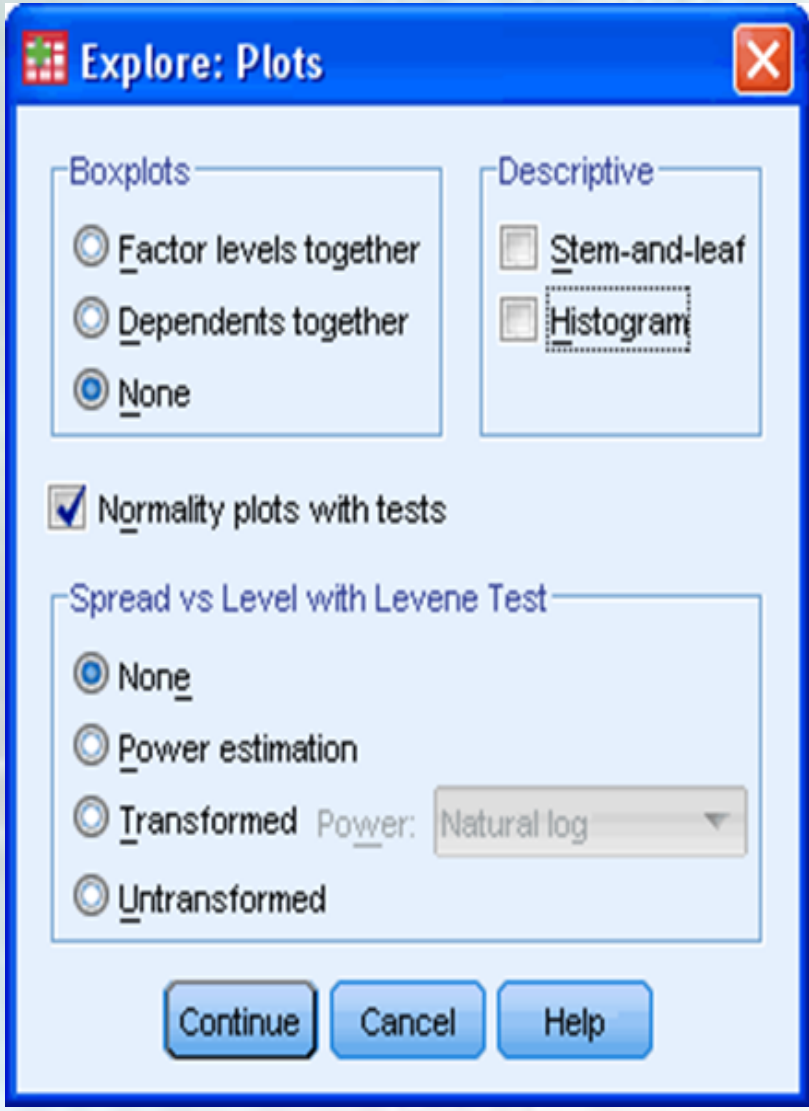
- Cochran's Q test is a non-parametric statistical test to verify if k treatments have identical effects where the response variable can take only two possible outcomes (coded as 0 and 1)

	Treatment 1	**Treatment 2**	**· · ·**	**Treatment *k***
Block 1	X_{11}	X_{12}	· · ·	X_{1k}
Block 2	X_{21}	X_{22}	· · ·	X_{2k}
Block 3	X_{31}	X_{32}	· · ·	X_{3k}
⋮	⋮	⋮	⋱	⋮
Block *b*	X_{b1}	X_{b2}	· · ·	X_{bk}

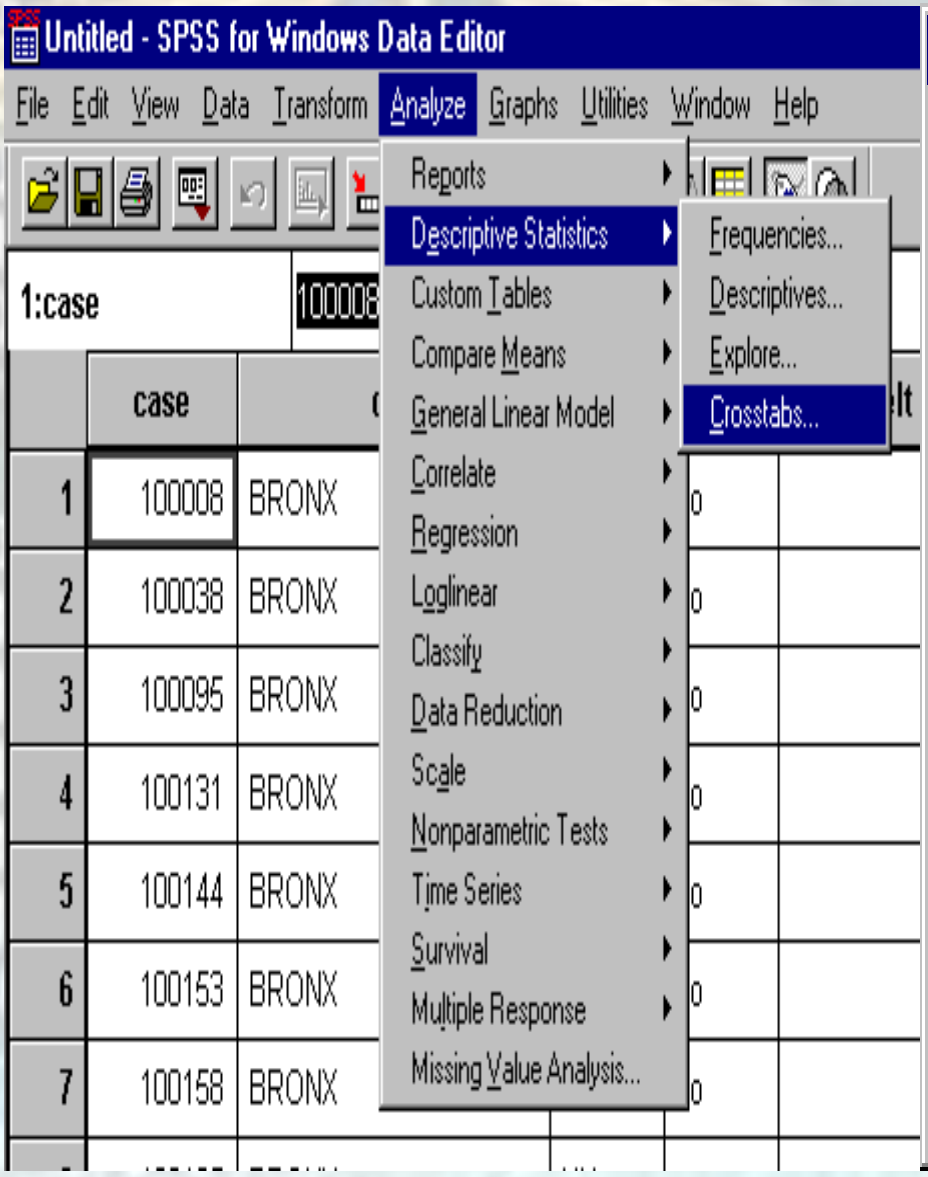
Applying the tests in SPSS software

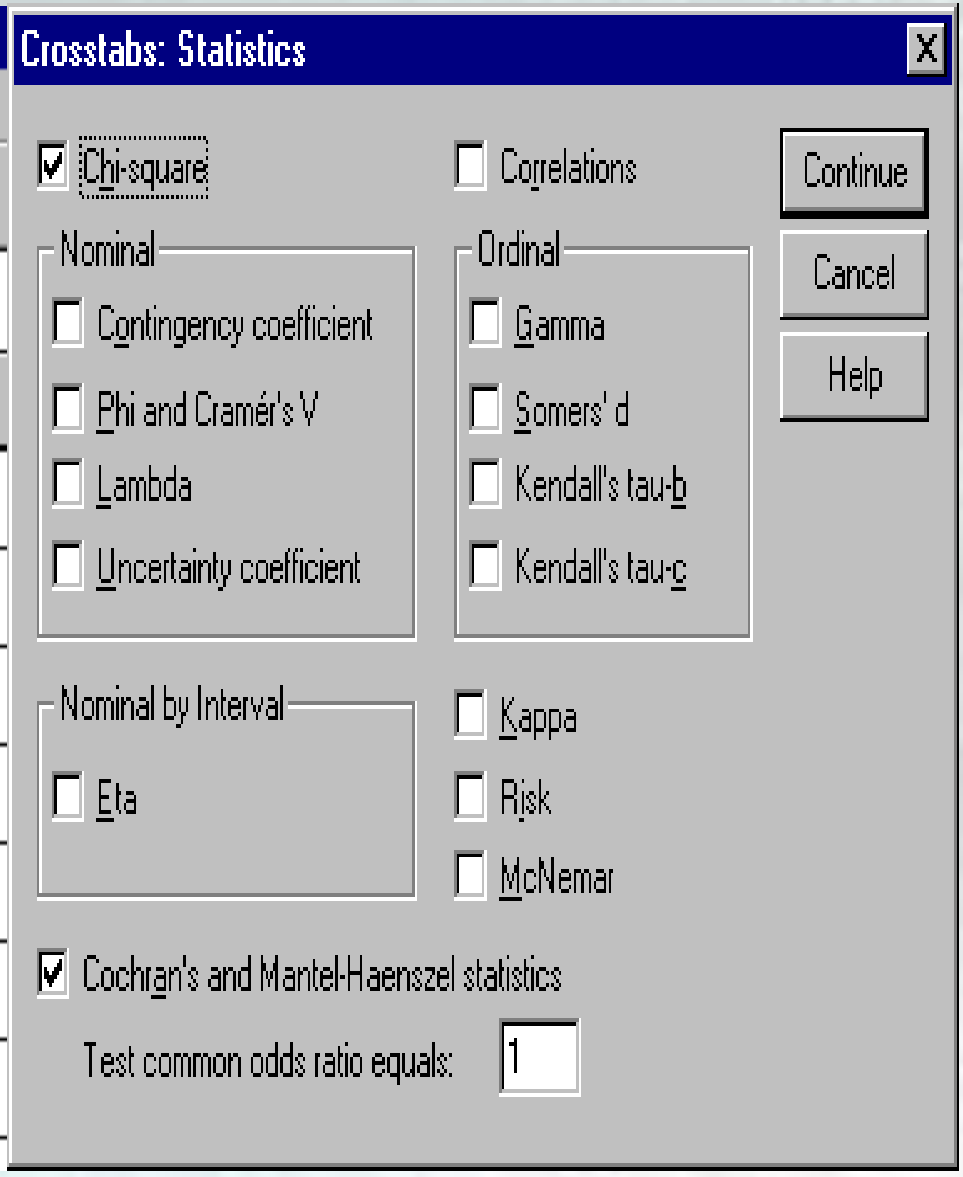
Normality tests



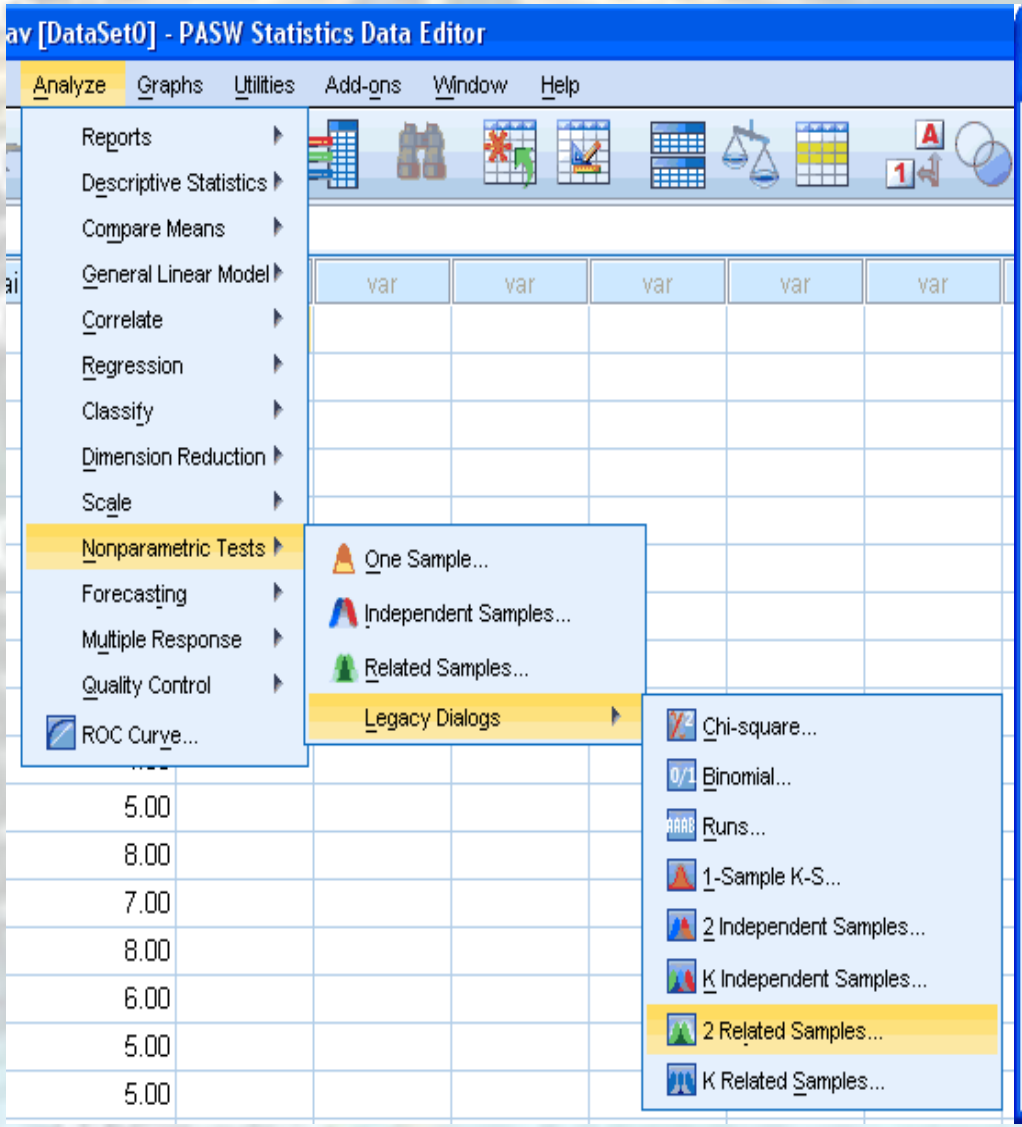


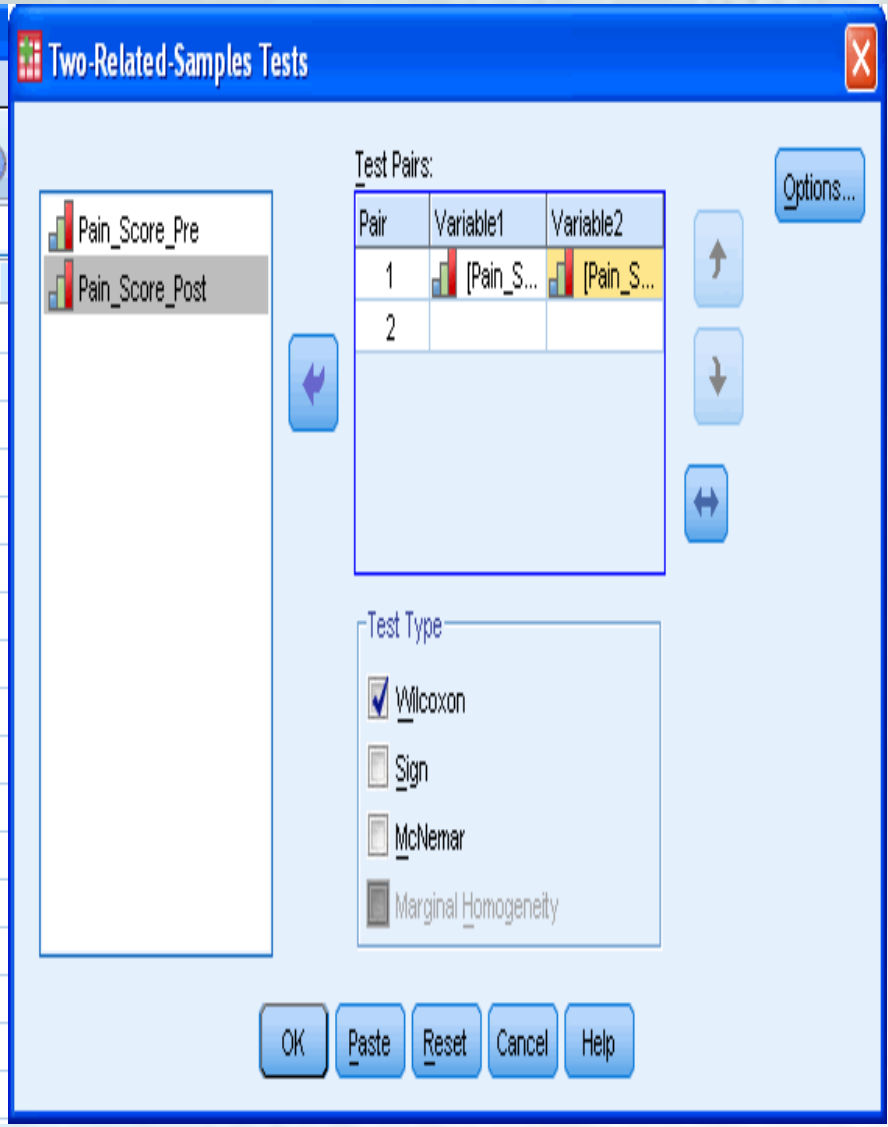
Chi-square tests



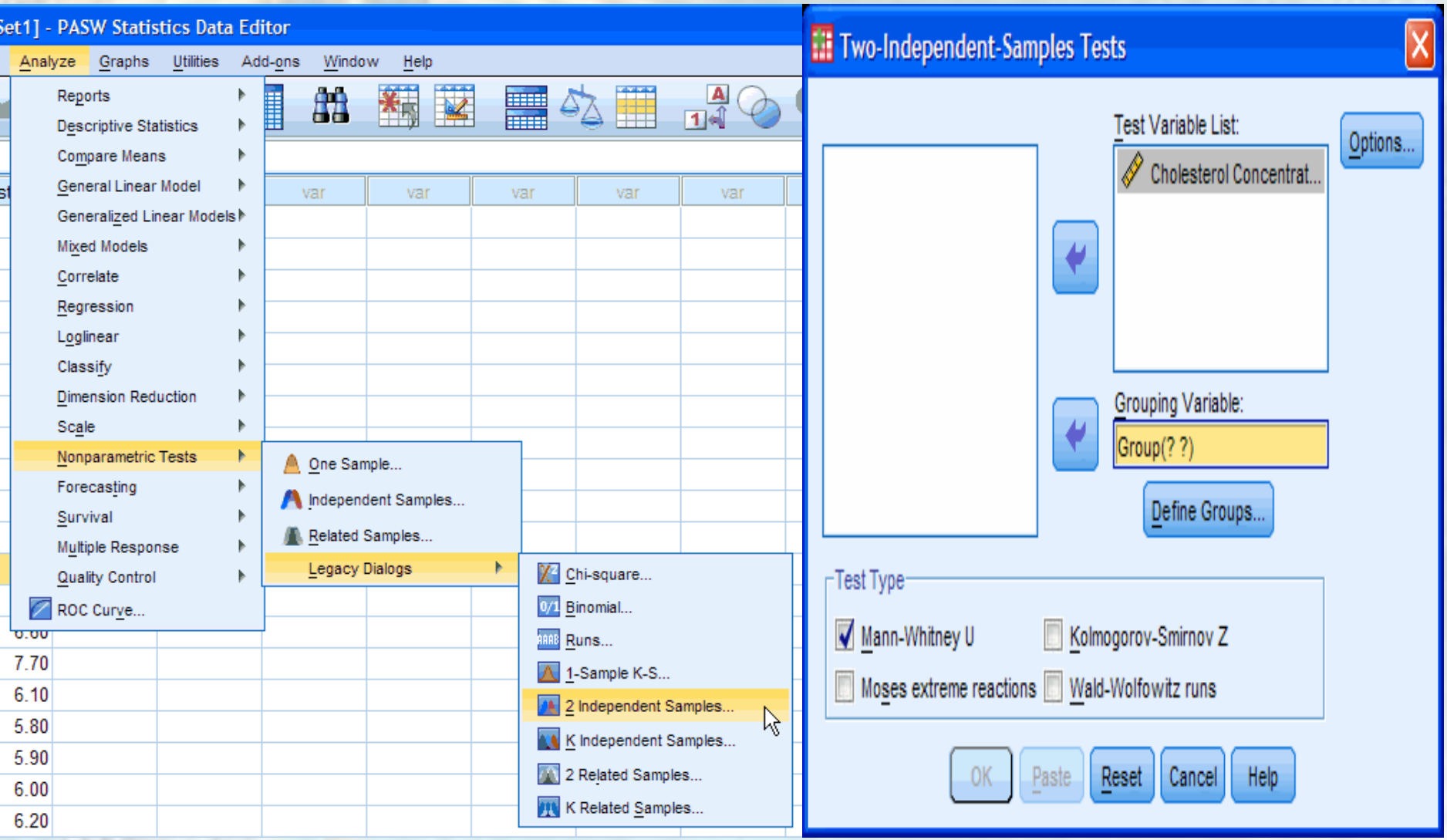


The Sign, Wilcoxon and McNemar test

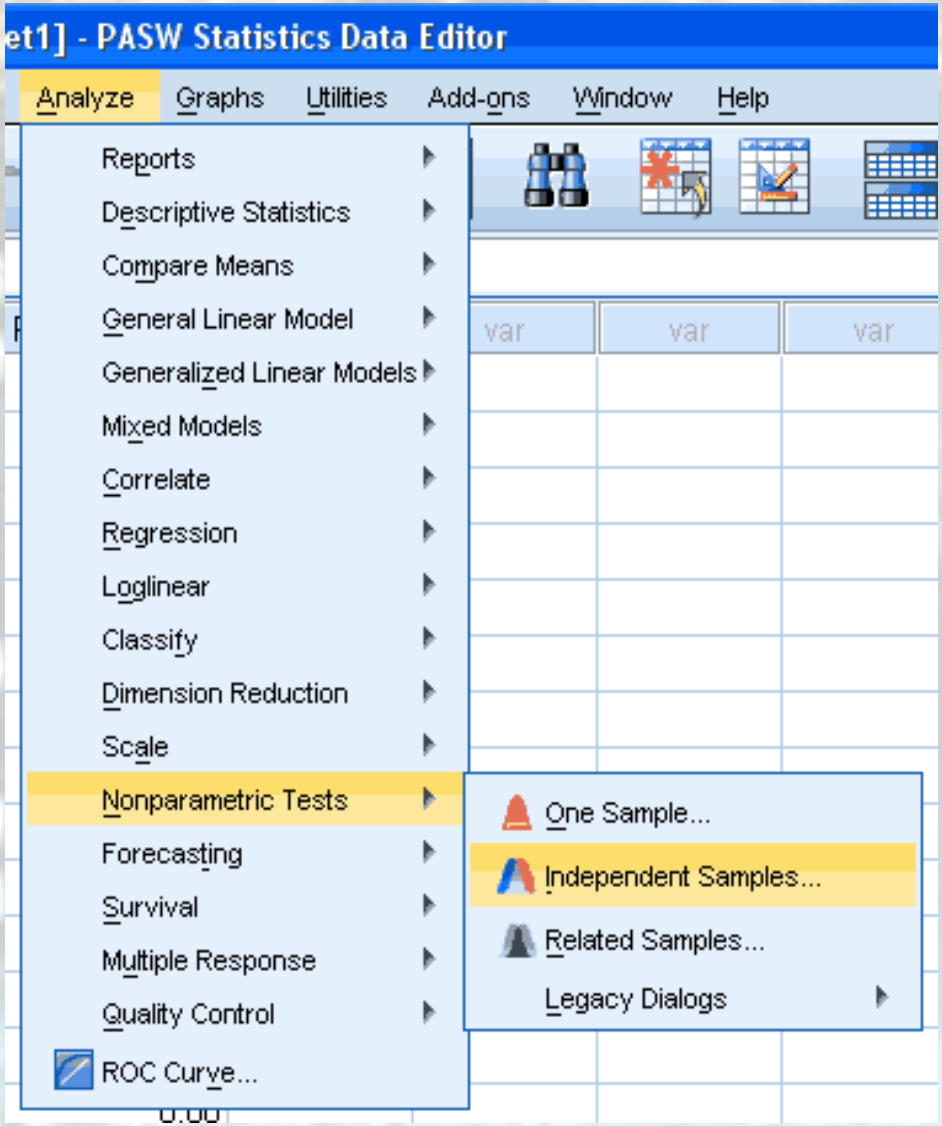


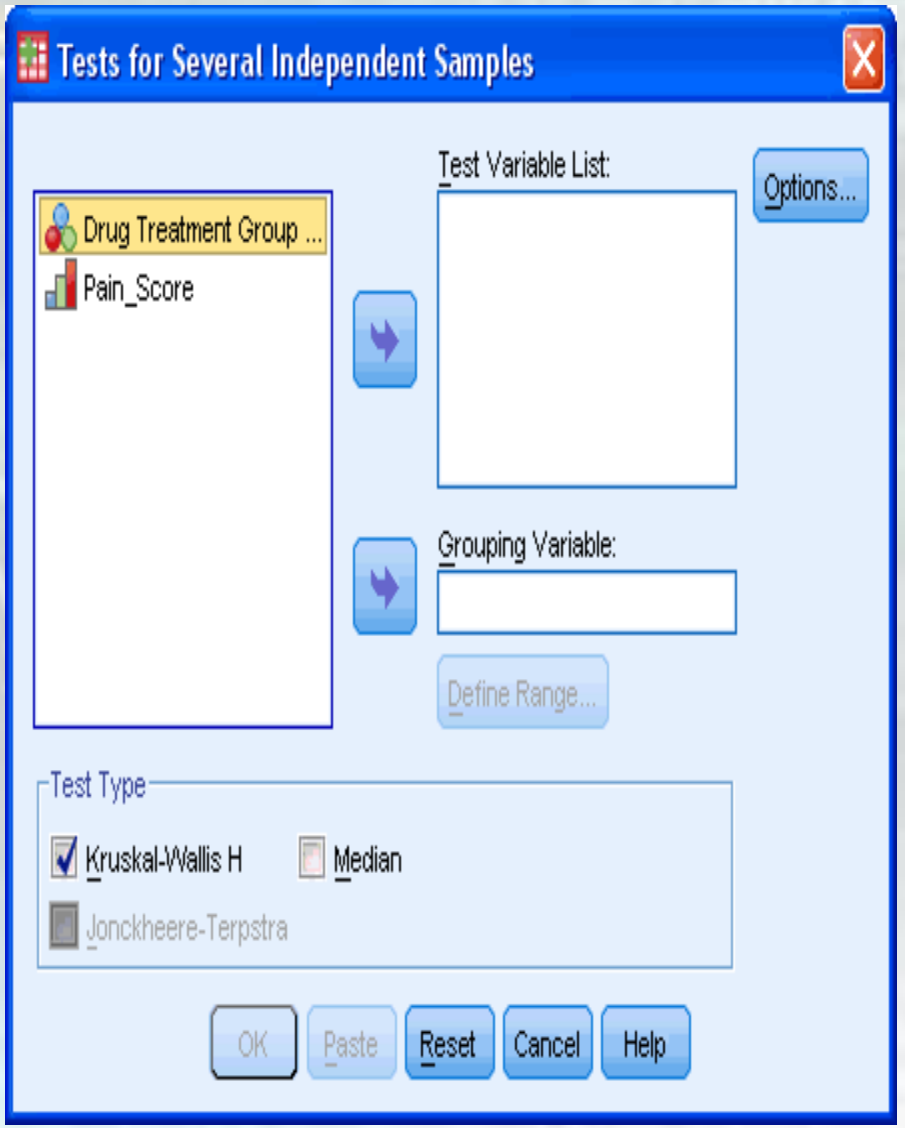


Mann Whitney U test

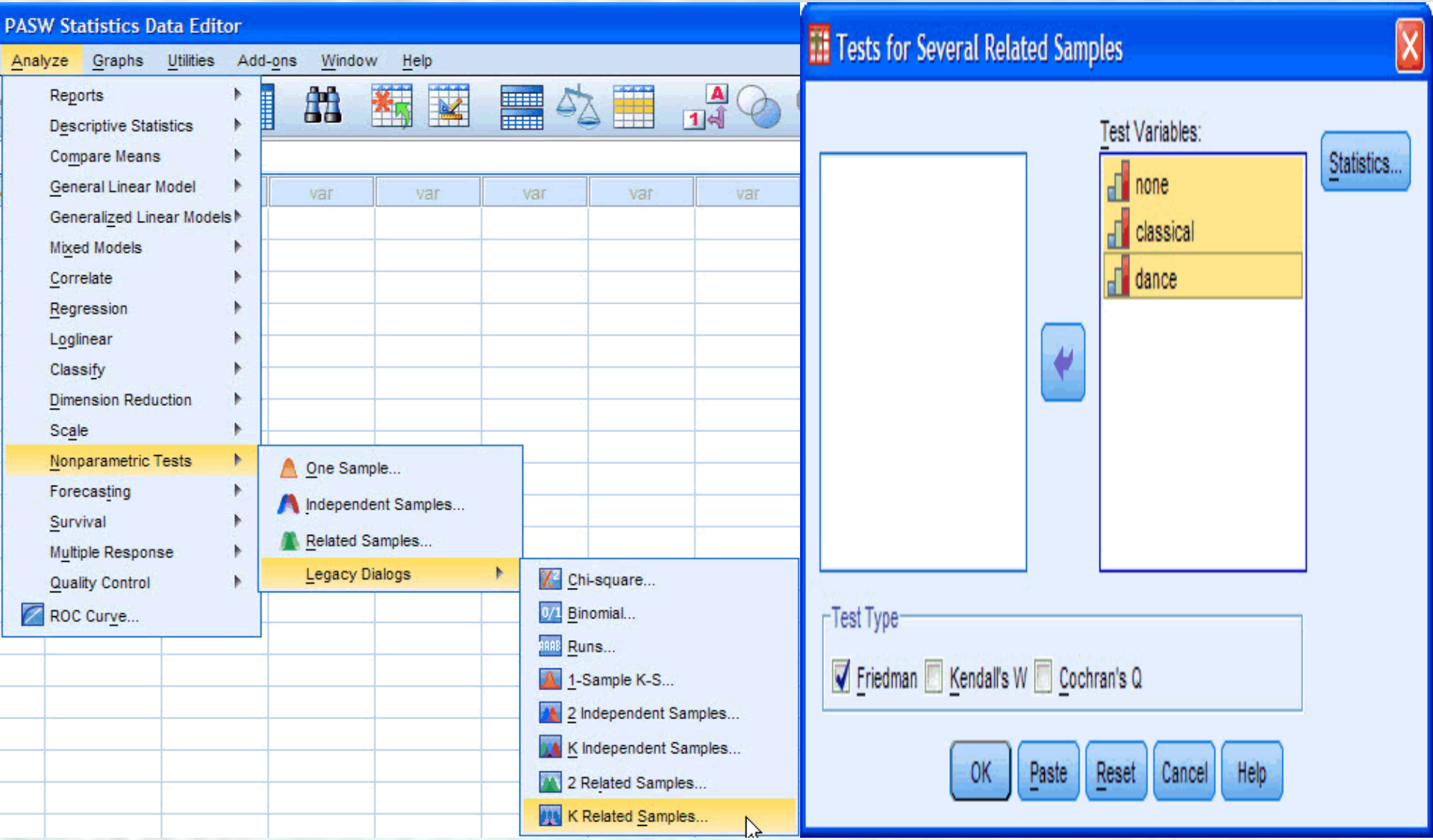


Kruskal Wallis H test

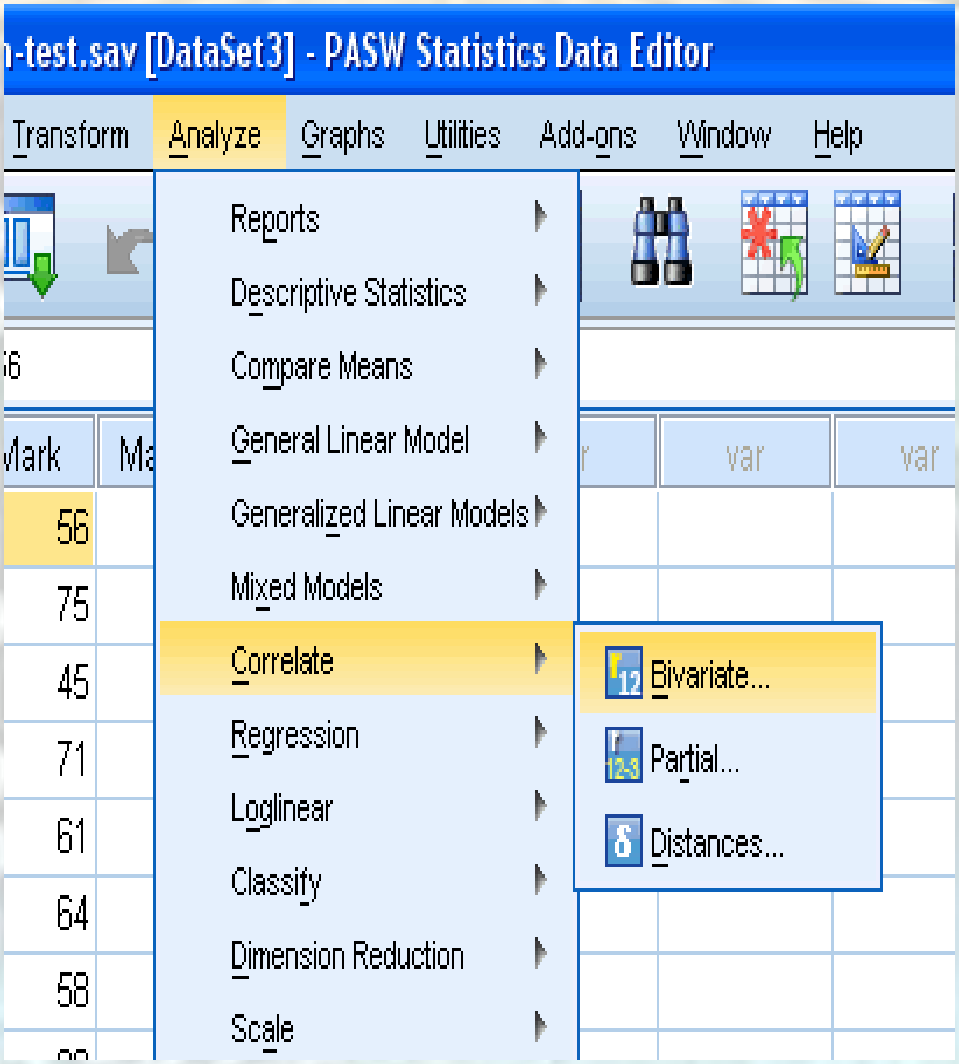


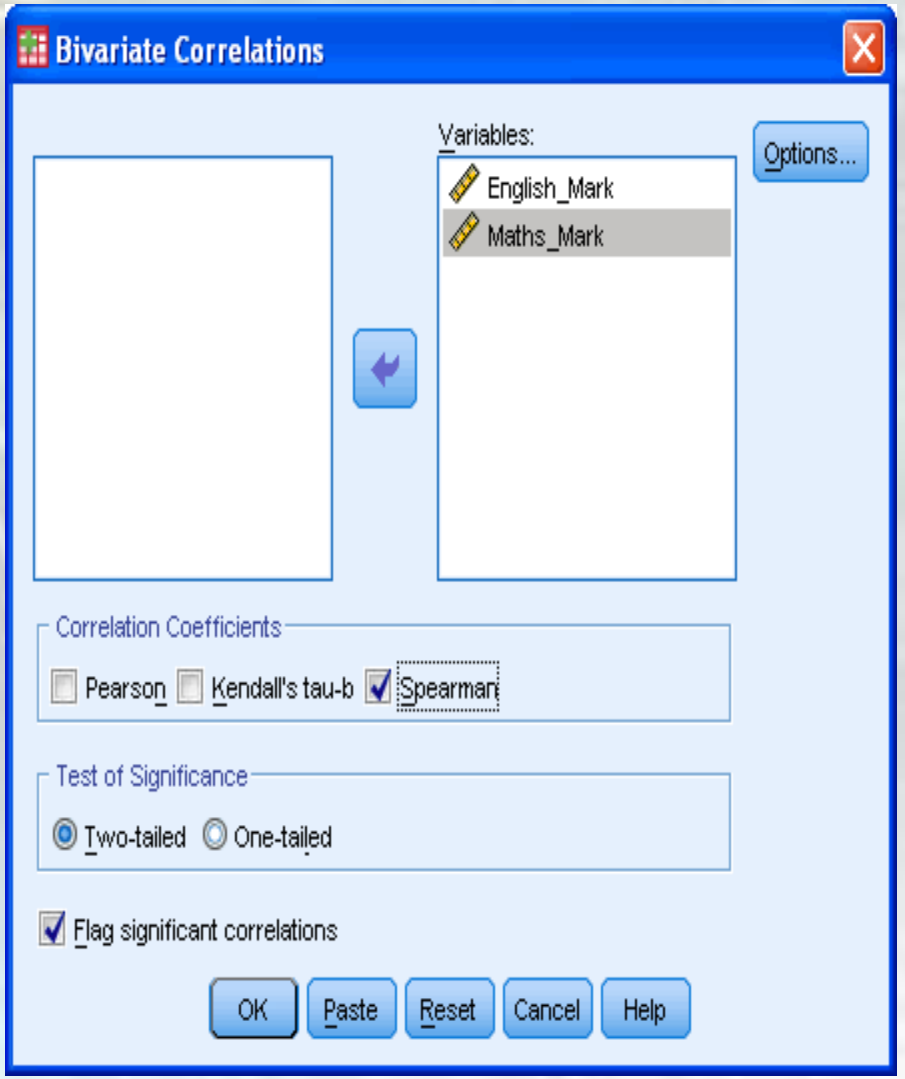


Friendman's ANOVA and Cochran's



Spearman's rho correlation test





Advantages of non-parametric tests

- These tests are distribution free.
- Easier to calculate & less time consuming than parametric tests when sample size is small.
- Can be used with any type of data.
- Many non-parametric methods make it possible to work with very small samples, particularly helpful in collecting pilot study data or medical researcher working with a rare disease.

Limitations of non-parametric methods

- Statistical methods which require no assumptions about populations are usually less efficient .
- As the sample size get larger , data manipulations required for non-parametric tests becomes laborious
- A collection of tabulated critical values for a variety of non-parametric tests under situations dealing with various sample sizes is not readily available.

Summary Table of Statistical Tests

Level of Measurement	Sample Characteristics					Correlation
	1 Sample	2 Sample		K Sample (i.e., >2)		
		Independent	Dependent	Independent	Dependent	
Categorical or Nominal	X^2	X^2	MacNemar test	X^2	Cochran's Q	
Rank or Ordinal	X^2	Mann Whitney U	Wilcoxon Signed Rank	Kruskal Wallis H	Friedman's ANOVA	Spearman's rho
Parametric (Interval & Ratio)	z test or t test	t test between groups	t test within groups	1 way ANOVA between groups	Repeated measure ANOVA	Pearson's test
		Factorial (2 way) ANOVA				


Thank You!